Agile Development Process and Devops Lab - (ISWE406P).

FAT EXAMINATION – SET 6

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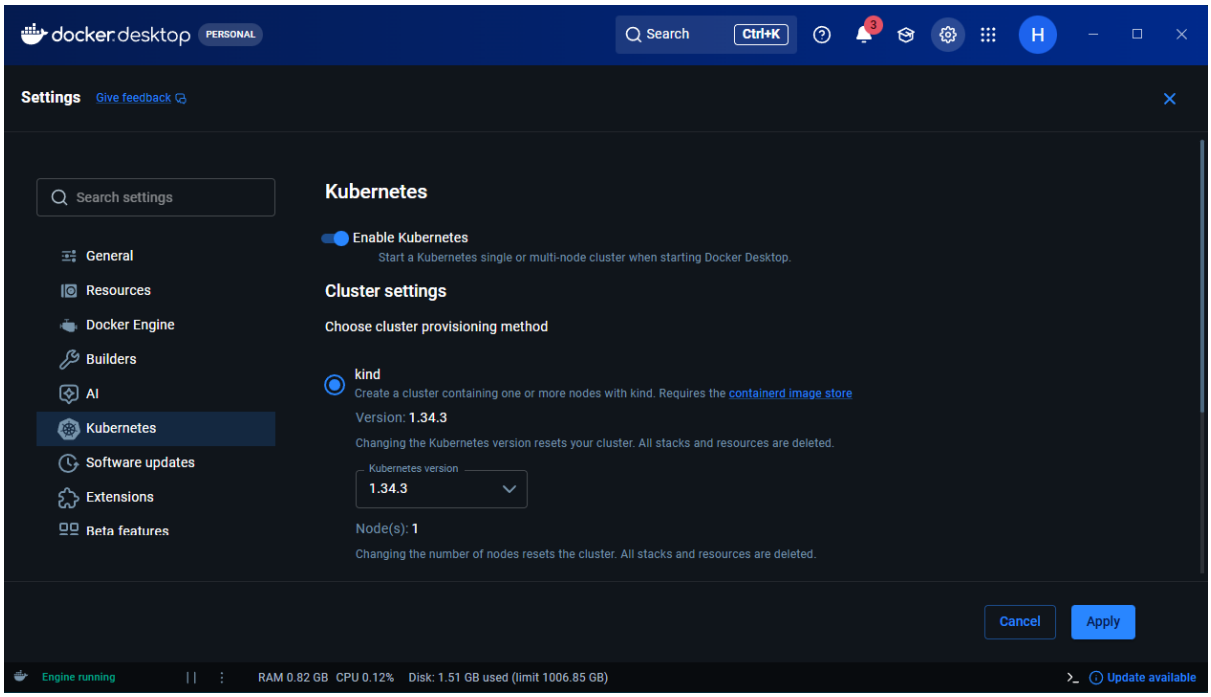
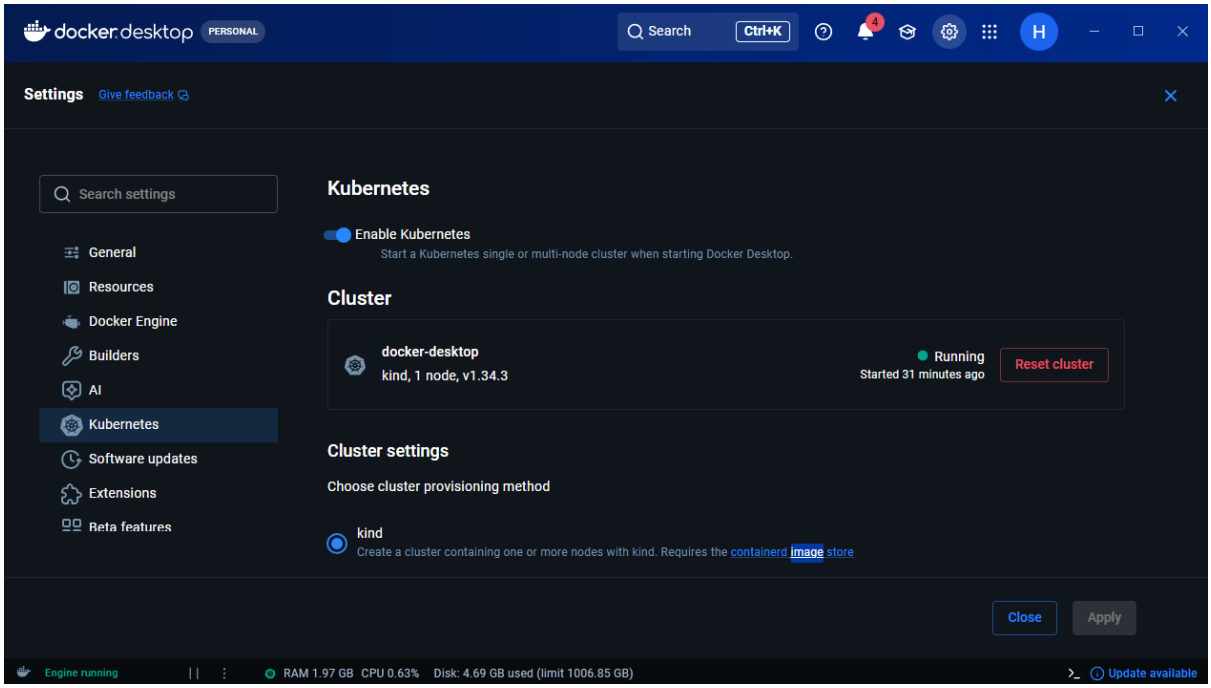
Problem Statement

Ensure user expenses are recorded and calculated correctly .

1: Install Docker Desktop

Install and enable Kubernetes, enable three plugins (Docker, Docker Pipeline, Kubernetes CLI) and install them





2: Create Java Application (in local machine)

```
expense-tracker-app
│
├── src
│   ├── main/java/com/expense/ExpenseService.java
│   └── test/java/com/expense/ExpenseServiceTest.java
│
├── pom.xml
├── Dockerfile
├── deployment.yaml
└── Jenkinsfile
```

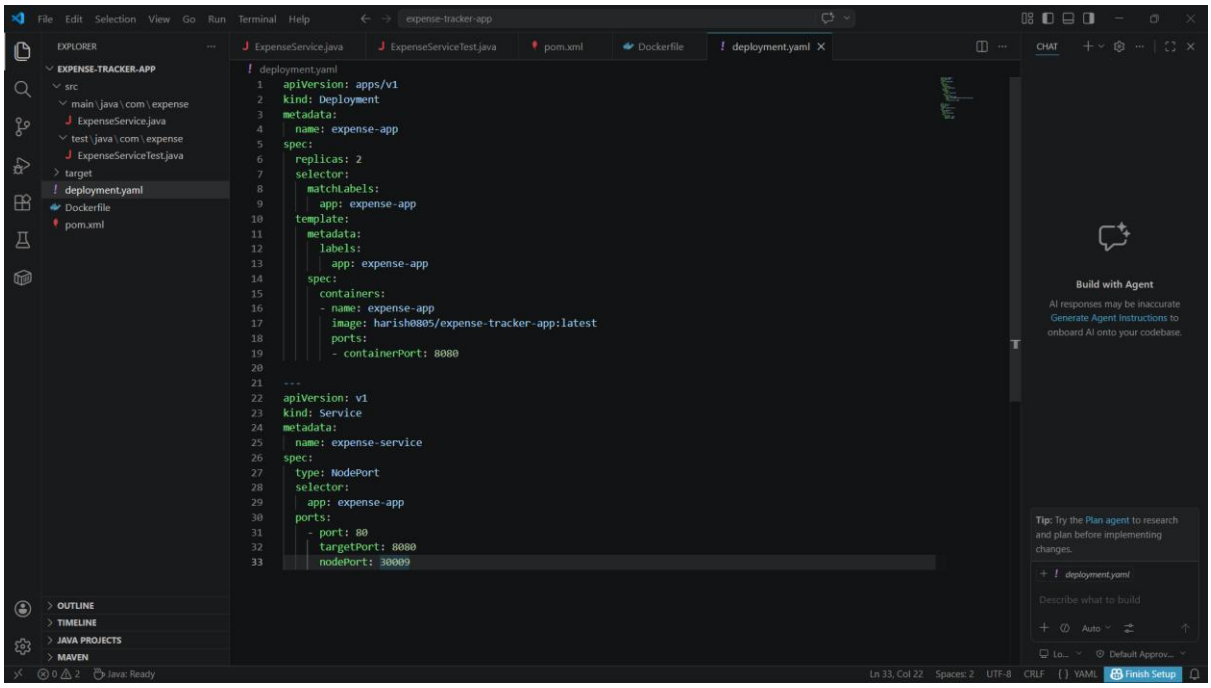
ExpenseService.java

```java
package com.expense;

import java.util.ArrayList;
import java.util.List;

public class ExpenseService {

    private List<Double> expenses = new ArrayList<>();

    // Add expense
    public boolean addExpense(double amount) {
        if (amount <= 0) {
            return false;
        }
        expenses.add(amount);
        return true;
    }

    // Calculate total
    public double getTotalExpense() {
        double total = 0;
        for (double e : expenses) {
            total += e;
        }
        return total;
    }

    public static void main(String[] args) {
        System.out.println("Expense Tracker Running...");
    }
}
```

ExpenseServiceTest.java

```java
package com.expense;

import static org.junit.jupiter.api.Assertions.*;
import org.junit.jupiter.api.Test;

public class ExpenseServiceTest {

    @Test
    void validExpenseTest() {
        ExpenseService service = new ExpenseService();
        assertTrue(service.addExpense(amount: 500));
    }

    @Test
    void invalidExpenseTest() {
        ExpenseService service = new ExpenseService();
        assertFalse(service.addExpense(-100));
    }

    @Test
    void totalCalculationTest() {
        ExpenseService service = new ExpenseService();
        service.addExpense(amount: 100);
        service.addExpense(amount: 200);
        assertEquals(300, service.getTotalExpense());
    }

    @Test
    void emptyTotalTest() {
        ExpenseService service = new ExpenseService();
        assertEquals(0, service.getTotalExpense());
    }
}
```

Deployment.yaml



```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: expense-app
spec:
  replicas: 2
  selector:
    matchLabels:
      app: expense-app
  template:
    metadata:
      labels:
        app: expense-app
    spec:
      containers:
      - name: expense-app
        image: harish0805/expense-tracker-app:latest
        ports:
        - containerPort: 8080

---
apiVersion: v1
kind: Service
metadata:
  name: expense-service
spec:
  type: NodePort
  selector:
    app: expense-app
  ports:
    - port: 80
      targetPort: 8080
      nodePort: 30009
```

Dockerfile

FROM eclipse-temurin:17-jdk-alpine

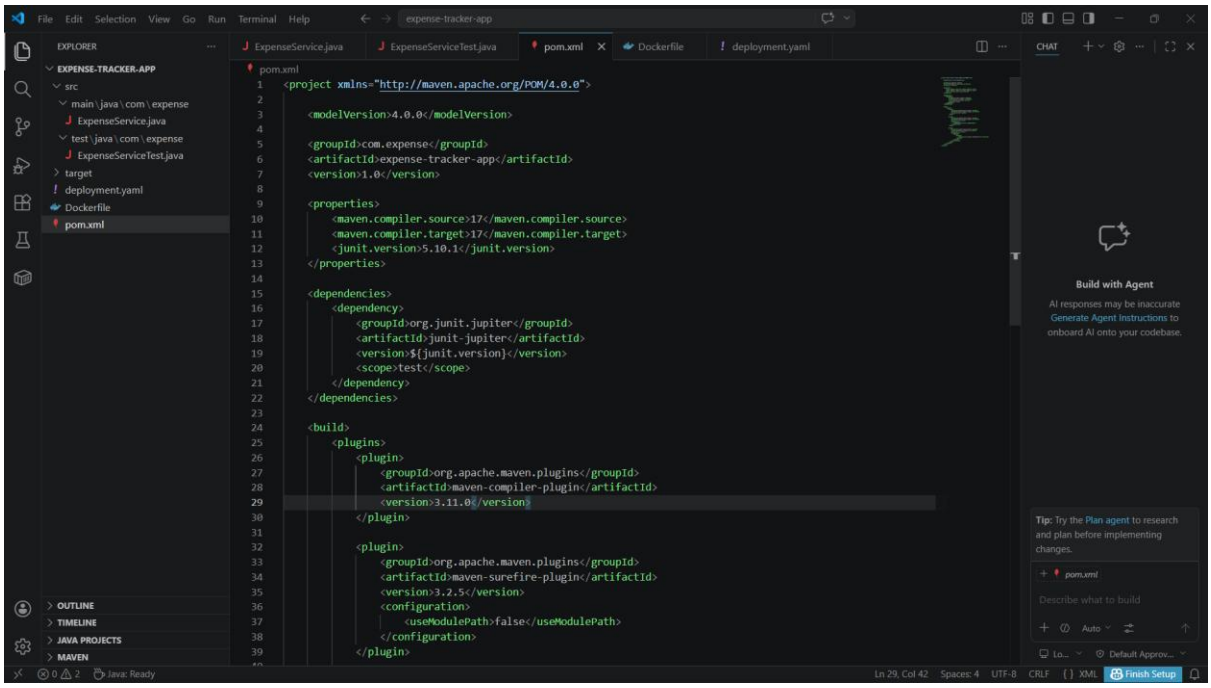
WORKDIR /app

COPY target/expense-tracker-app-1.0.jar app.jar

EXPOSE 8080

ENTRYPOINT ["java", "-jar", "app.jar"]

Pom.xml

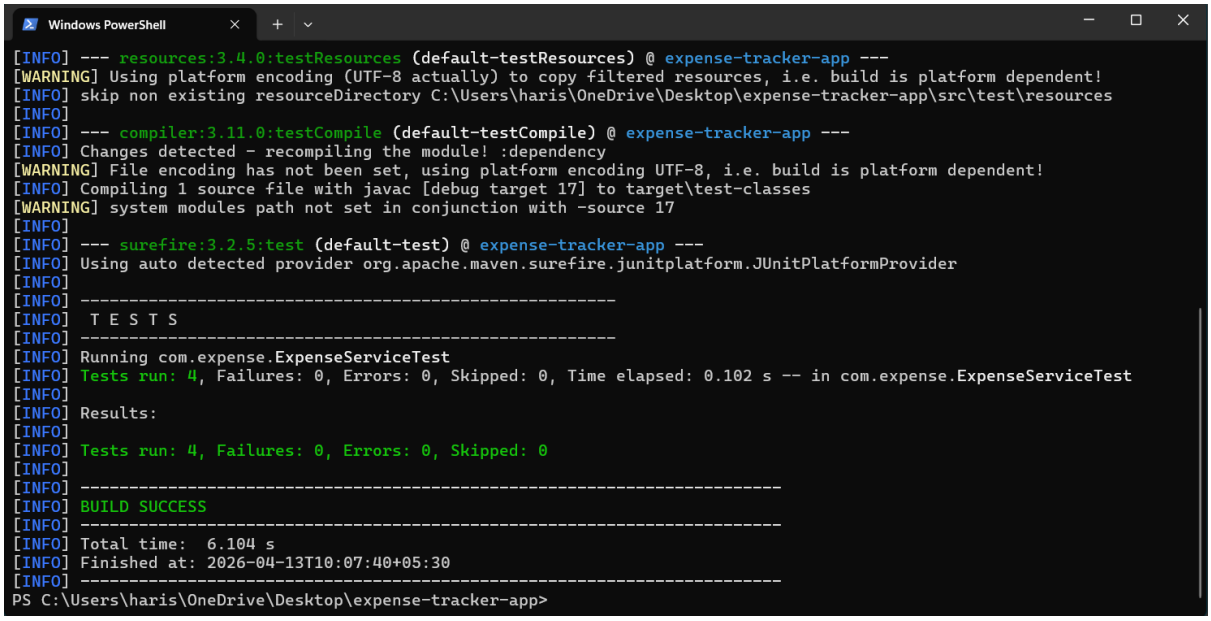


```xml
<project xmlns="http://maven.apache.org/POM/4.0.0">

    <modelVersion>4.0.0</modelVersion>

    <groupId>com.expense</groupId>
    <artifactId>expense-tracker-app</artifactId>
    <version>1.0</version>

    <properties>
        <maven.compiler.source>17</maven.compiler.source>
        <maven.compiler.target>17</maven.compiler.target>
        <junit.version>5.10.1</junit.version>
    </properties>

    <dependencies>
        <dependency>
            <groupId>org.junit.jupiter</groupId>
            <artifactId>junit-jupiter</artifactId>
            <version>${junit.version}</version>
            <scope>test</scope>
        </dependency>
    </dependencies>

    <build>
        <plugins>
            <plugin>
                <groupId>org.apache.maven.plugins</groupId>
                <artifactId>maven-compiler-plugin</artifactId>
                <version>3.11.0</version>
            </plugin>

            <plugin>
                <groupId>org.apache.maven.plugins</groupId>
                <artifactId>maven-surefire-plugin</artifactId>
                <version>3.2.5</version>
                <configuration>
                    <useModulePath>false</useModulePath>
                </configuration>
            </plugin>
```

3. Build Locally using Maven (Test)

mvn clean test

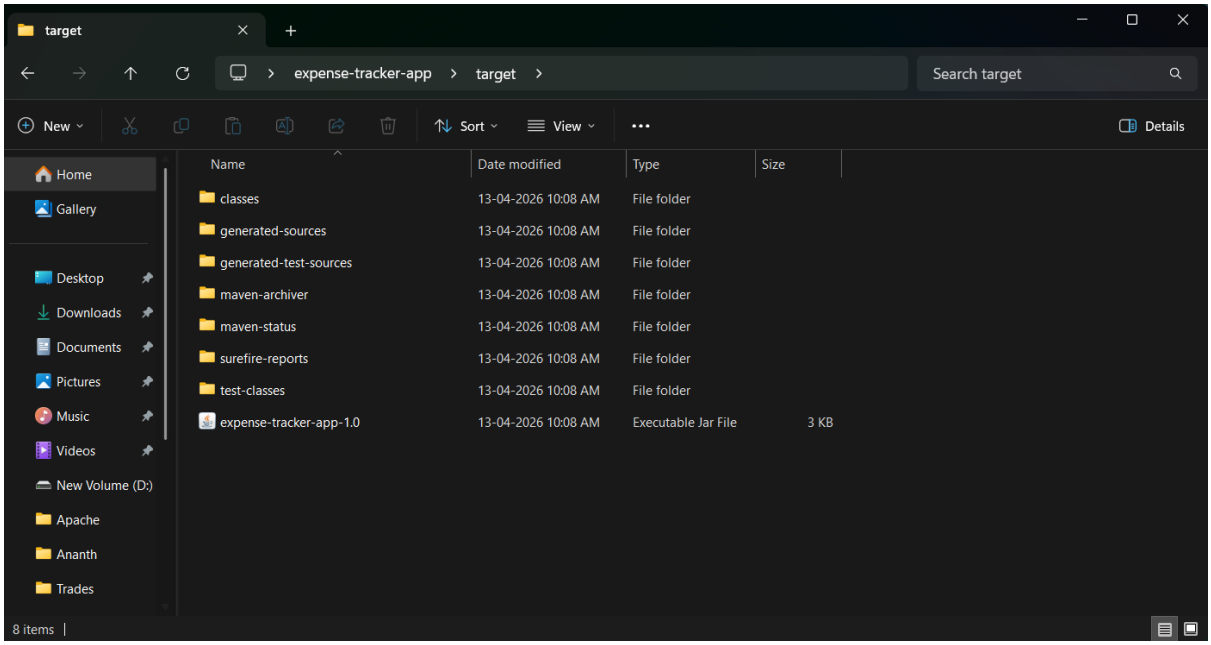
mvn clean package



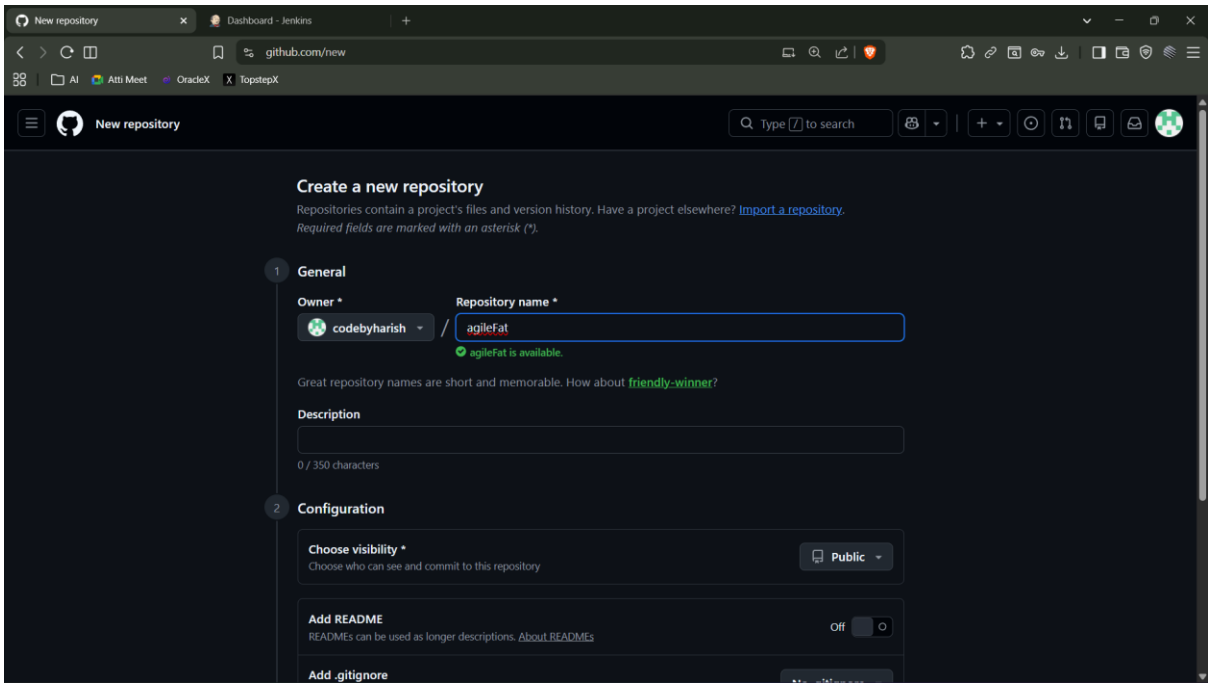
```
[INFO] --- resources:3.4.0:testResources (default-testResources) @ expense-tracker-app ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory C:\Users\haris\OneDrive\Desktop\expense-tracker-app\src\test\resources
[INFO]
[INFO] --- compiler:3.11.0:testCompile (default-testCompile) @ expense-tracker-app ---
[INFO] Changes detected - recompiling the module! :dependency
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 1 source file with javac [debug target 17] to target\test-classes
[WARNING] system modules path not set in conjunction with -source 17
[INFO]
[INFO] --- surefire:3.2.5:test (default-test) @ expense-tracker-app ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running com.expense.ExpenseServiceTest
[INFO] Tests run: 4, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.102 s -- in com.expense.ExpenseServiceTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 4, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  6.104 s
[INFO] Finished at: 2026-04-13T10:07:40+05:30
[INFO] ------------------------------------------------------------------------
PS C:\Users\haris\OneDrive\Desktop\expense-tracker-app>
```

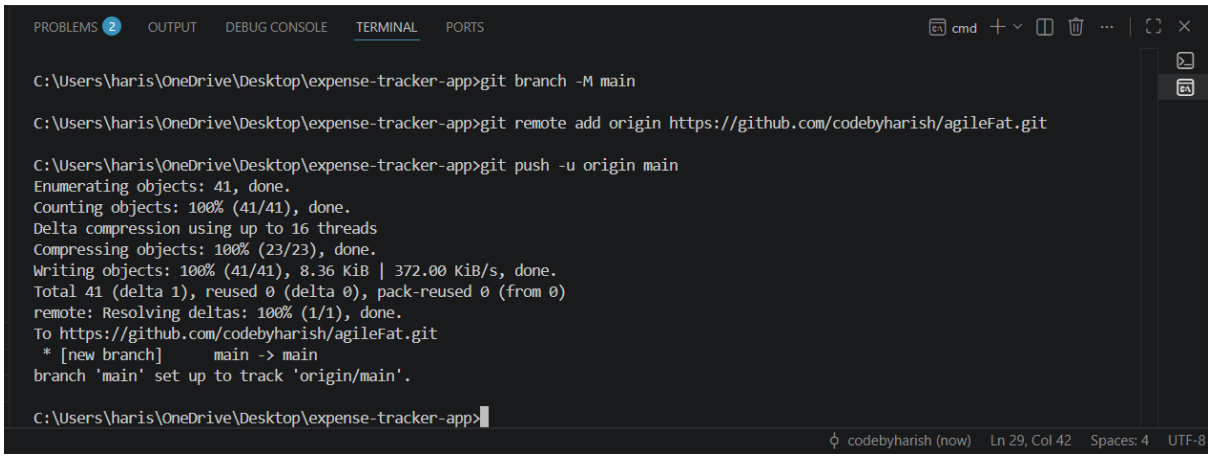
```
Windows PowerShell
[INFO]
[INFO] --- compiler:3.11.0:testCompile (default-testCompile) @ expense-tracker-app ---
[INFO] Changes detected - recompiling the module! :dependency
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 1 source file with javac [debug target 17] to target\test-classes
[WARNING] system modules path not set in conjunction with -source 17
[INFO]
[INFO] --- surefire:3.2.5:test (default-test) @ expense-tracker-app ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running com.expense.ExpenseServiceTest
[INFO] Tests run: 4, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.089 s -- in com.expense.ExpenseServiceTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 4, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
[INFO] --- jar:3.4.1:jar (default-jar) @ expense-tracker-app ---
[INFO] Building jar: C:\Users\haris\OneDrive\Desktop\expense-tracker-app\target\expense-tracker-app-1.0.jar
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  8.699 s
[INFO] Finished at: 2026-04-13T10:08:20+05:30
[INFO] ------------------------------------------------------------------------
PS C:\Users\haris\OneDrive\Desktop\expense-tracker-app>
```

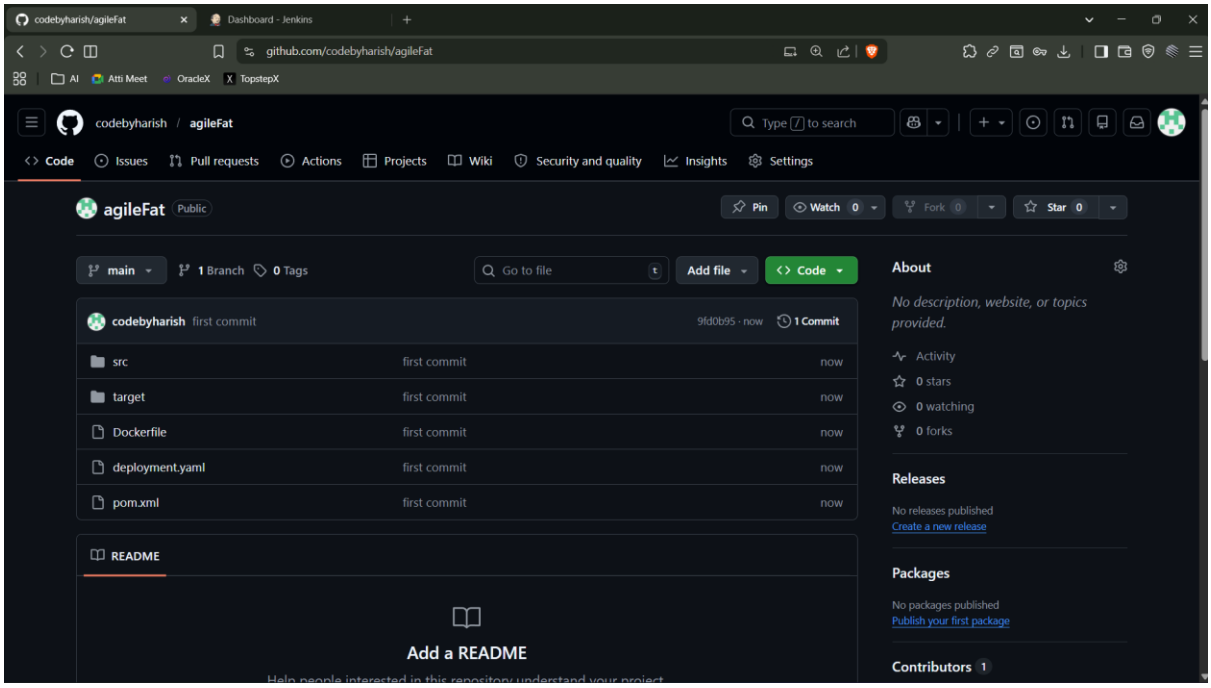
target/ test-java-app-1.0.jar



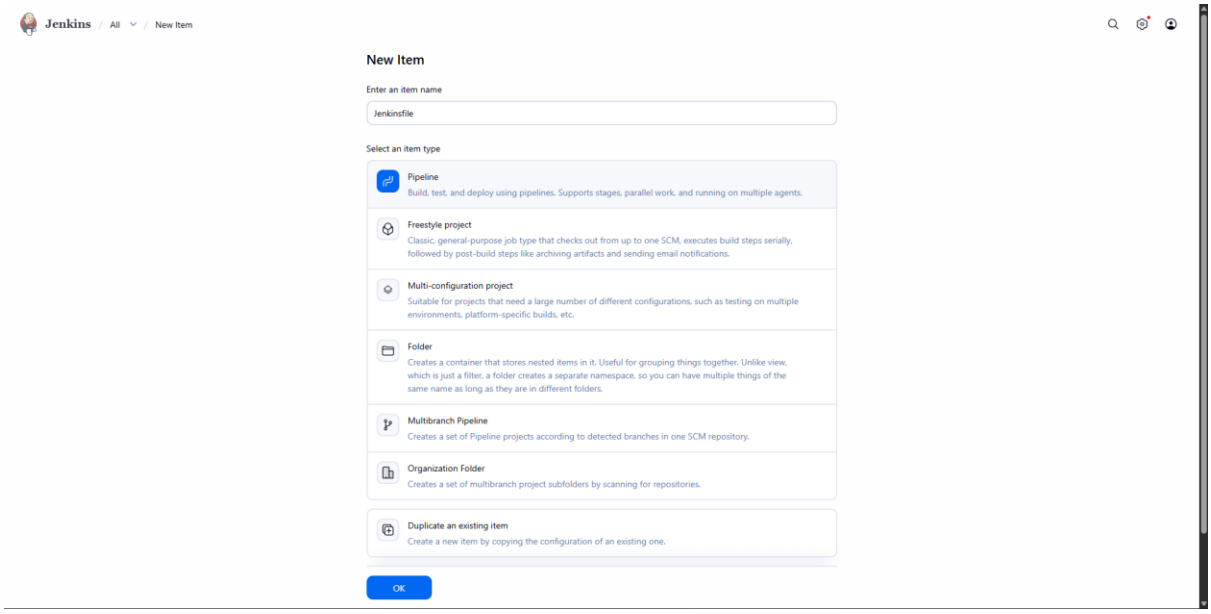
4. Push Code to GitHub

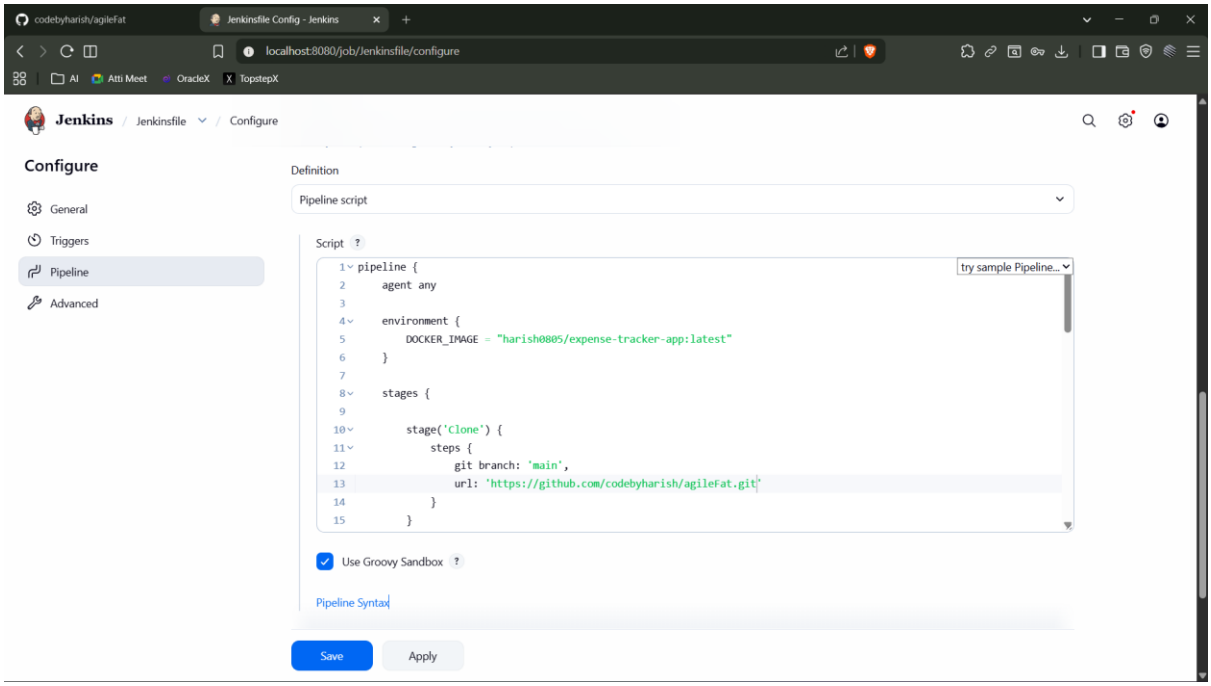






5. Create Jenkins Pipeline Job





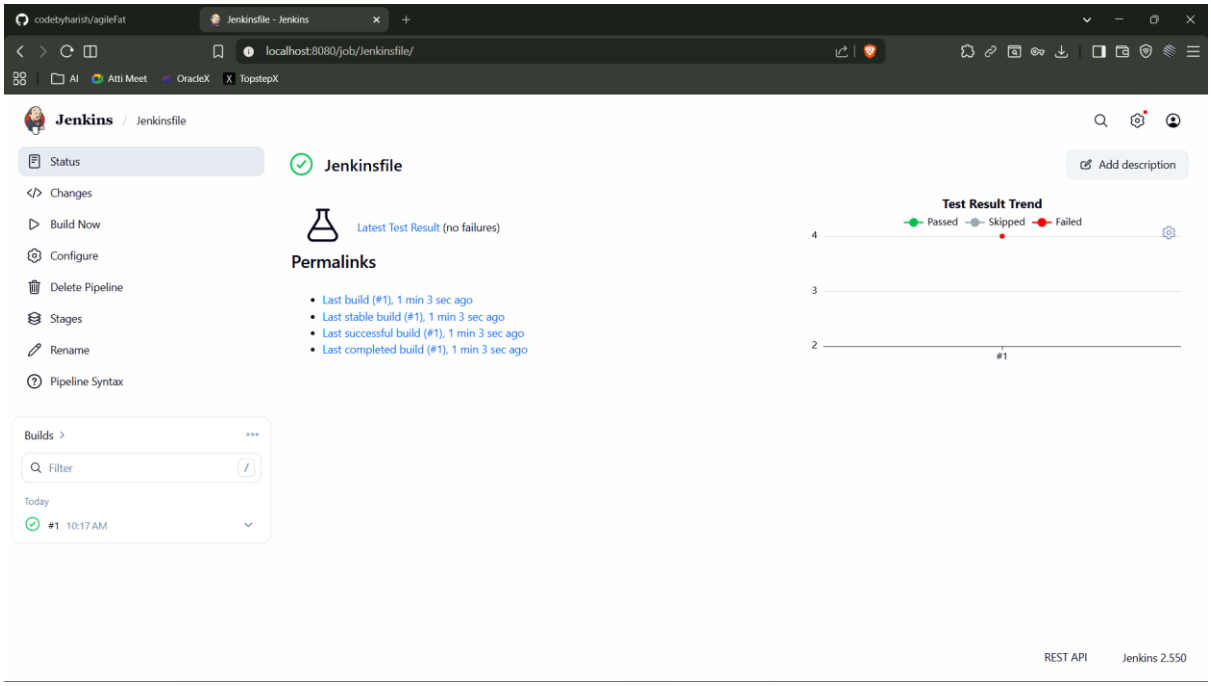
Pipeline Code

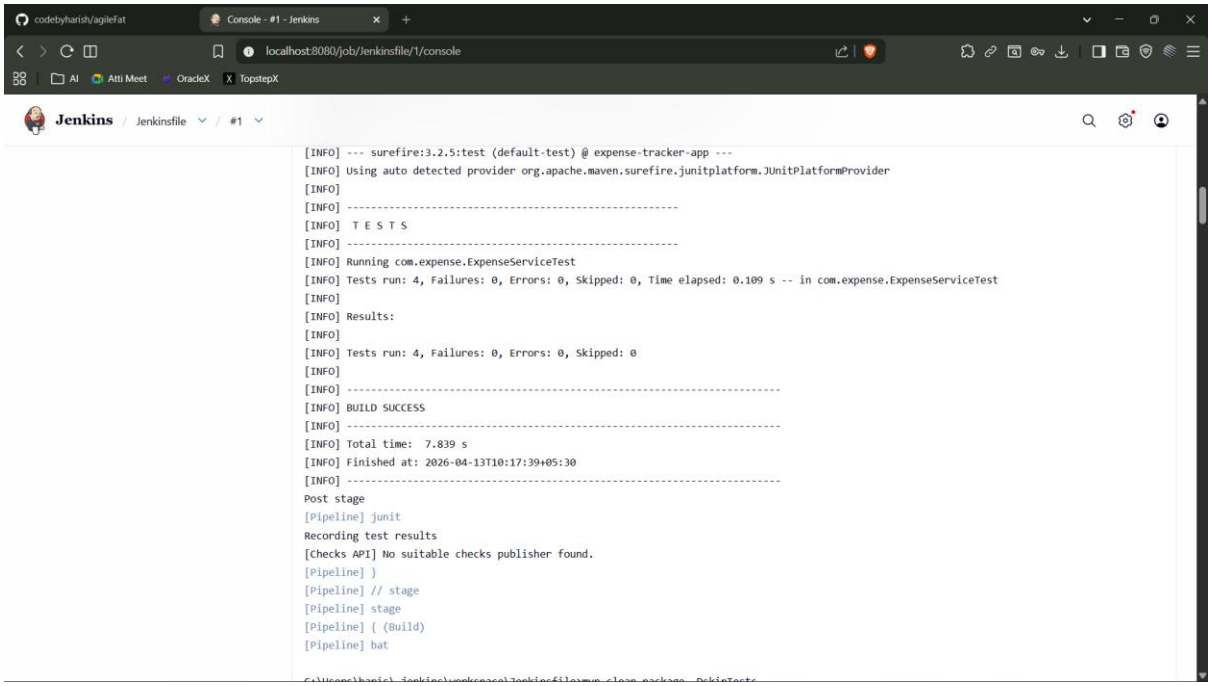
```
pipeline {
  agent any
  environment {
    DOCKER_IMAGE = "harish0805/expense-tracker-app:latest"
  }
  stages {
    stage('Clone') {
      steps {
        git branch: 'main',
        url: 'https://github.com/codebyharish/agileFat.git'
      }
    }
    stage('Test') {
```

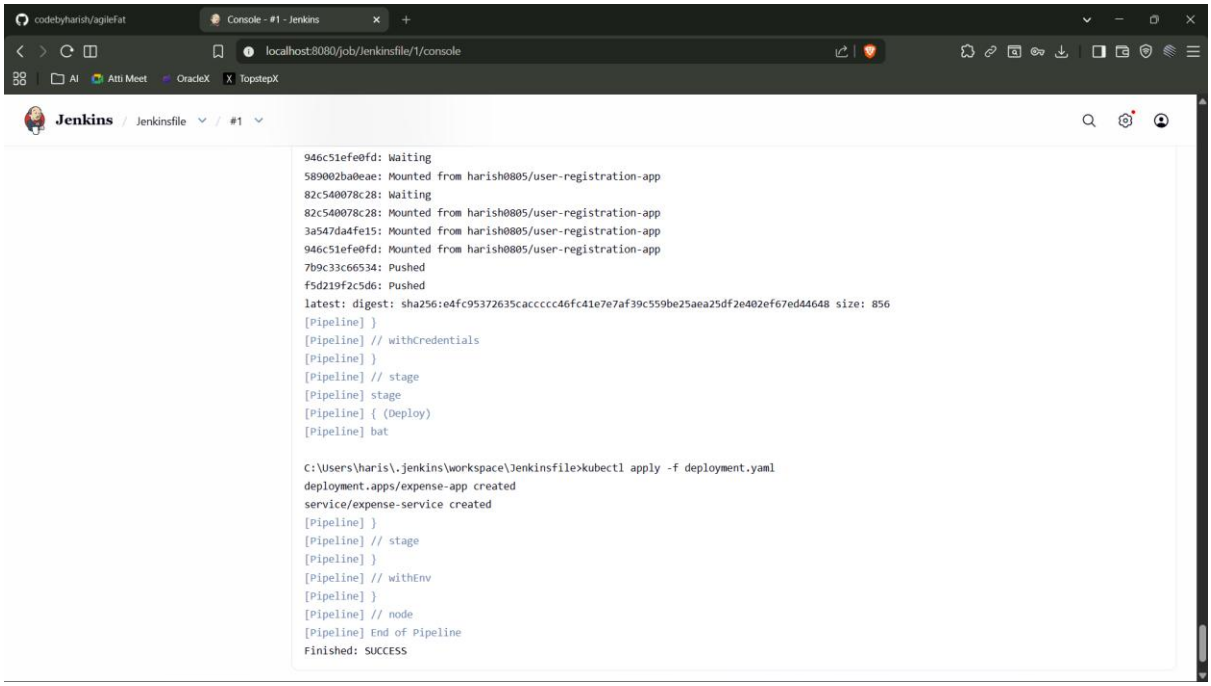
```
    steps {
        bat 'mvn clean test'
    }
    post {
        always {
            junit 'target/surefire-reports/*.xml'
        }
    }
}
stage('Build') {
    steps {
        bat 'mvn clean package -DskipTests'
    }
}
stage('Docker Build') {
    steps {
        bat 'docker build -t %DOCKER_IMAGE% .'
    }
}
stage('Docker Push') {
    steps {
        withCredentials([usernamePassword(
            credentialsId: 'dockerhub-creds',
            usernameVariable: 'DOCKER_USER',
            passwordVariable: 'DOCKER_PASS'
        )]) {
            bat """
            docker login -u %DOCKER_USER% -p %DOCKER_PASS%
```

```
            docker push %DOCKER_IMAGE%
            """
        }
    }
  }
  stage('Deploy') {
    steps {
      bat 'kubectl apply -f deployment.yaml'
    }
  }
 }
}
```

Build the Pipeline



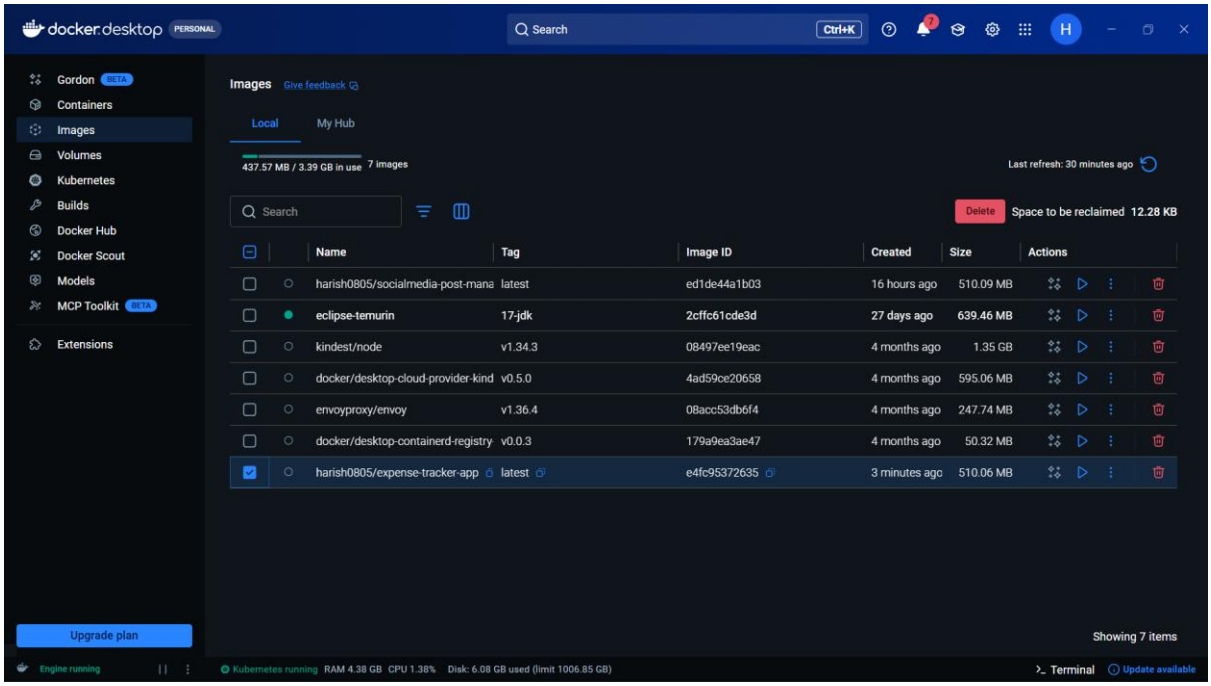


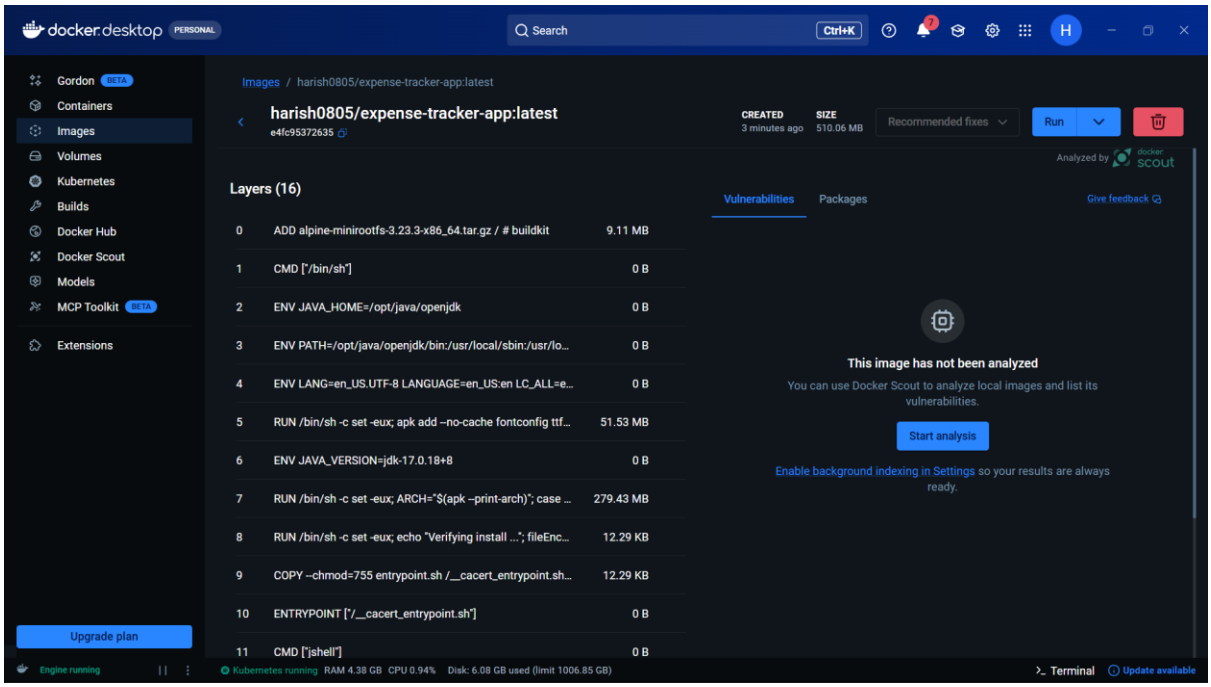


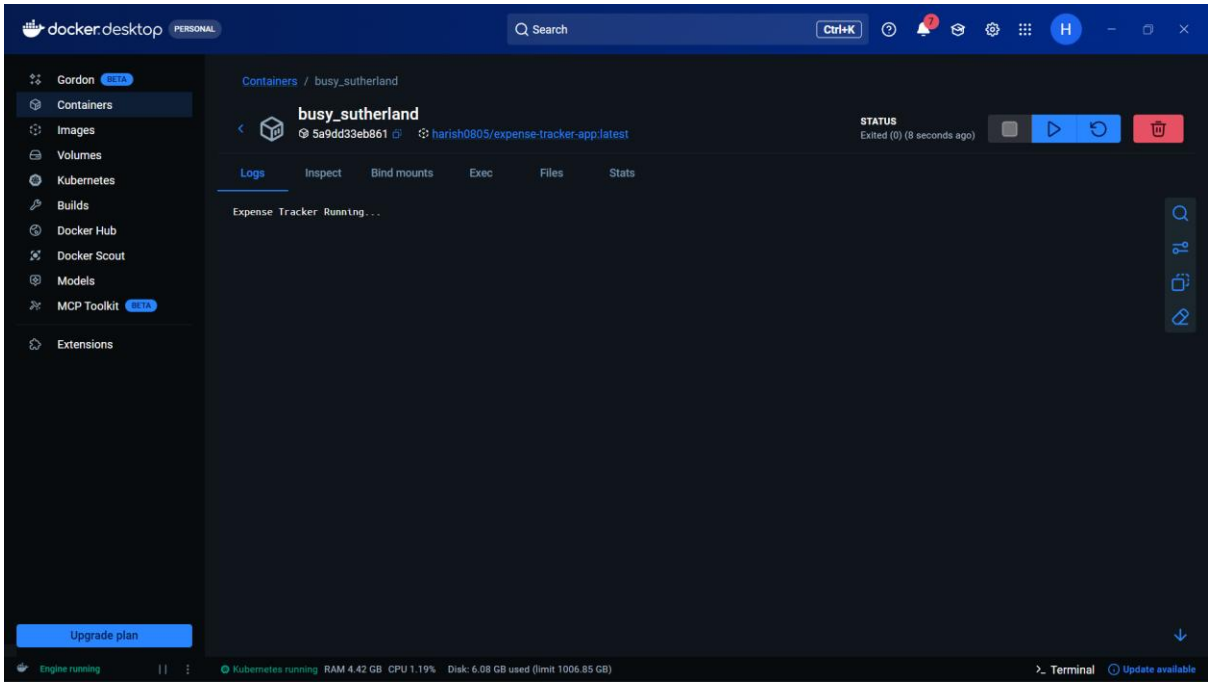
Above console output is the success message

6. Verify Docker and Kubernetes Deployment

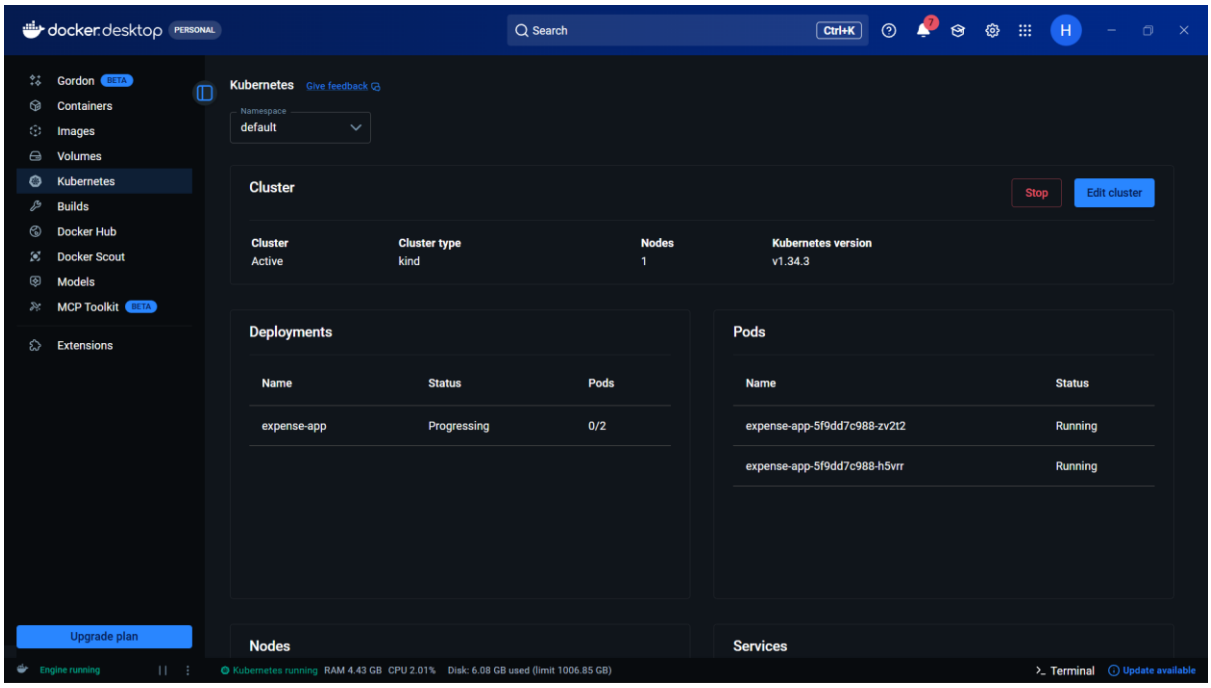
Docker Image






docker desktop
PERSONAL
Search
Ctrl+K
Gordon
Containers
Images
Volumes
Kubernetes
Builds
Docker Hub
Docker Scout
Models
MCP Toolkit
Extensions
Upgrade plan
Containers / busy_sutherland
busy_sutherland
5a9dd33eb861
harish0805/expense-tracker-app:latest
STATUS
Exited (0) (8 seconds ago)
Logs
Inspect
Bind mounts
Exec
Files
Stats
Expense Tracker Running...
Engine running
Kubernetes running RAM 4.42 GB CPU 1.19% Disk: 6.08 GB used (limit 1006.85 GB)
Terminal
Update available

Kubernetes


docker desktop
PERSONAL
Search
Ctrl+K
Gordon
Containers
Images
Volumes
Kubernetes
Builds
Docker Hub
Docker Scout
Models
MCP Toolkit
Extensions
Upgrade plan
Kubernetes
Give feedback
Namespace
default
Cluster
Stop
Edit cluster
Cluster
Active
Cluster type
kind
Nodes
1
Kubernetes version
v1.34.3
Deployments
Name
Status
Pods
expense-app
Progressing
0/2
Pods
Name
Status
expense-app-5f9dd7c988-zv2t2
Running
expense-app-5f9dd7c988-h5vrr
Running
Nodes
Services
Engine running
Kubernetes running RAM 4.43 GB CPU 2.01% Disk: 6.08 GB used (limit 1006.85 GB)
Terminal
Update available

```
Windows PowerShell
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Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\haris\OneDrive\Desktop\expense-tracker-app> kubectl get svc
NAME              TYPE        CLUSTER-IP     EXTERNAL-IP   PORT(S)        AGE
expense-service   NodePort    10.96.35.149   <none>        80:30009/TCP   4m23s
kubernetes        ClusterIP   10.96.0.1      <none>        443/TCP        26m
PS C:\Users\haris\OneDrive\Desktop\expense-tracker-app>
```